Yu Wang

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https://mrgodfrey.github.io

Research Profile

My research concerns fundamental questions in control theory and inverse problems for partial differential equations. I am particularly interested in how controllability, observability, unique continuation, and identifiability are preserved, weakened, or lost under randomness, discretization, and partial observation.

A major part of my work studies the transfer of control and observability mechanisms across different mathematical settings. This includes understanding when deterministic control structures can be lifted to stochastic systems, and when continuous estimates and unique continuation properties remain valid, uniformly and quantitatively, after numerical discretization.

On the inverse-problem side, I study what can be recovered from limited or indirect observations, with emphasis on identifiability and quantitative stability for unknown sources and related structures. More broadly, I am interested in the mathematical boundary between what information a PDE system retains and what is irretrievably lost.

Research interests: Control Theory; Inverse Problems; Stochastic PDEs

Appointments

Assistant Professor 06.2023 - present
School of Mathematics, Southwest Jiaotong University

Visiting Scholar 09.2025 - 02.2026
Chair for Dynamics, Control, Machine Learning and Numerics, FAU Erlangen-Nürnberg

Host Professor: Prof. Dr. DhC. Enrique Zuazua

Education

Ph.D. in Mathematics 09.2017 - 06.2023
Sichuan University

Supervisor: Professor Qi Lü

B.Sc. in Mathematics 09.2013 - 06.2017
Sichuan University

Publications

Books

1. Inverse Problems for Stochastic Partial Differential Equations

Q. Lü, **Y. Wang**

SpringerBriefs on PDEs and Data Science, Springer Nature Singapore, 2026. ISBN 9789819590476 Book | ArXiv | Blog

Journal Articles

1. Exact controllability for stochastic first-order multi-dimensional hyperbolic systems

Z. Li, Q. Lü, **Y. Wang**, H. Yang

SIAM J. Control Optim., Accepted ArXiv | Blog

2. The ϕ-null controllability for stochastic fourth order semi-discrete parabolic equations

Y. Wang, Q. Zhao

Appl. Math. Optim., 95 (2027), Paper No. 3 Article | ArXiv | Blog

3. scDCL: A multi-view single-cell RNA sequencing clustering method based on dual contrastive learning

L. Gan, H. Meng, Y. Chen, **Y. Wang**

Computational Biology and Chemistry, (2026), 108998 Article | Blog

4. An Inverse Source Problem for Semilinear Stochastic Hyperbolic Equations

Q. Lü, **Y. Wang**

Inverse Problems, 41(11) (2025), 115014 Article | ArXiv | Blog

5. The ϕ-null controllability for semi-discrete stochastic semilinear parabolic equations

Y. Wang, Q. Zhao

ESAIM Control Optim. Calc. Var., 31 (2025), Paper No. 98 Article | ArXiv | Blog

6. Exact controllability for a refined stochastic plate equation

Q. Lü, **Y. Wang**

Chinese Ann. Math. Ser. B, 46(3) (2025), 415-442 Article | ArXiv | Blog

7. Stability and regularization for ill-posed Cauchy problem of a stochastic parabolic differential equation

F. Dou, P. Lü, **Y. Wang**

Inverse Problems, 40(11) (2024), 115005 Article | ArXiv | Blog

8. Null controllability for stochastic coupled systems of fourth order parabolic equations

Y. Wang

J. Math. Anal. Appl., 538 (2024), 128426 Article | ArXiv | Blog

9. Null controllability for fourth order stochastic parabolic equations

Q. Lü, **Y. Wang**

SIAM J. Control Optim., 60 (2022), 1563-1590 Article | ArXiv | Blog

Preprints

1. A Transfer Principle for Elliptic Carleman Estimates: Application to Discrete Calderón Stability

Q. Lü, **Y. Wang**

Preprint ArXiv

2. A Unified Kalman-Type Classification of Controllability for Linear Stochastic Systems

Q. Lü, **Y. Wang**

Submitted ArXiv | Lean | Blog

3. From Continuous to Fully Discrete Carleman Estimates: A Transfer Principle for Parabolic Schemes

Q. Lü, **Y. Wang**

Preprint ArXiv | Blog

4. Null Controllability of a Stochastic Parabolic Equation with a Space-Dependent Analytic Noise Coefficient

Q. Lü, **Y. Wang**

Submitted ArXiv | Blog

5. Stability Estimates for the Inverse Recovery of Drift and Diffusion Point Sources in Stochastic Parabolic Equations

Q. Lü, **Y. Wang**

Submitted ArXiv | Blog

6. Carleman Estimates for Wave Equations on the Half-Line: AI-Assisted Weights and Applications

C. An, **Y. Wang**, Q. Ye, Z. Zhang, Q. Zhuang

Submitted ArXiv | Blog

7. A Geometric Inverse Source Problem for Stochastic Parabolic Equations

Y. Li, Q. Lü, M. Qian, **Y. Wang**

Submitted ArXiv | Blog

Research Grants

Controllability of stochastic plate equations 2026-2027
Sichuan Science and Technology Program (No. 2026NSFSC0777)

Funding: RMB 100K

Controllability of fourth order semilinear stochastic partial differential equations 2025-2027
National Natural Science Foundation of China (No. 12401589)

Funding: RMB 300K

Controllability of a fourth-order coupled stochastic parabolic system 2024-2025
Fundamental Research Funds for the Central Universities (No. 2682024CX013)

Funding: RMB 100K

Teaching

Neural Networks and Deep Learning 2025-2026

Introduction to Major 2024

Academic Service

Mathematical Reviews / MathSciNet (American Mathematical Society) - reviewer

Journal refereeing

Applied Mathematics & Optimization; Asian Journal of Control; Chinese Annals of Mathematics, Series B; ESAIM: Control, Optimisation and Calculus of Variations; IMA Journal of Mathematical Control and Information; Inverse Problems; Journal of Differential Equations; Journal of Mathematical Analysis and Applications; Journal of Optimization Theory and Applications; Journal of Systems Science and Complexity; Mathematical Control and Related Fields; Neural Networks; Science China Mathematics; SIAM Journal on Control and Optimization; Systems & Control Letters.